

Reviewer Pack — Minimal Rank of Hodge Decompositions vs Permutation Circuit ...

Entry 1567143d861e · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Hodge Decompositions vs Permutation Circuit Depth

Entry ID: 1567143d861e **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-25 09:59:47 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (Hodge Theory) **Field B** (complexity object): Complexity Theory: Permutation Circuit Complexity

Statement:

{‘part1’: ‘For every satisfiable k -CNF formula F with n variables, the minimal rank of its associated Hodge decomposition is proportional to the depth of a permutation circuit that computes the truth table of F .’, ‘part2’: ‘Specifically, there exists a constant $c > 0$ such that for all satisfiable k -CNF formulas, the following inequality holds:’, ‘part3’: ‘ $\text{rank}(H(F)) \leq c \times \text{depth}(P_F)$ ’, ‘where’: {‘ $H(F)$ ’: ‘is the Hodge decomposition of the ideal generated by the polynomials encoding F ’, ‘ $\text{rank}(H(F))$ ’: ‘is the rank of the Hodge bundle associated with $H(F)$ ’, ‘ $\text{depth}(P_F)$ ’: ‘is the depth of a permutation circuit that computes the truth table of F ’}}

Rationale (proposer’s reasoning):

{‘part1’: ‘Hodge theory studies algebraic varieties and their cohomology, which can reveal geometric properties. By connecting it to permutation circuits, we aim to expose a geometric interpretation of computational complexity.’, ‘part2’: ‘If the conjecture holds, it would provide a new method for lower bounding the complexity of functions computed by permutation circuits.’, ‘part3’: ‘The relationship between Hodge theory and computational complexity could lead to novel algorithms or insights into the structure of computational problems.’}

Taxonomy category: HODGE_THEORY_PERMUTATION_CIRCUIT (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): b912924b4a127522

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the Spearman rank correlation coefficient (ρ) between the minimal rank of the Hodge decomposition and the depth of the permutation circuit is ≥ 0.8 for all tested k-CNF formulas, with no seed producing a $\rho < 0.6$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	SAFE	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Hodge decomposition" AND "permutation circuit depth" - "minimal rank" AND "k-CNF formula" AND Hodge - "truth table computation" AND circuit complexity AND Hodge bundle

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_kcnf(n, alpha):
        num_clauses = int(alpha * n * (n - 1) / 2)
        clauses = []
        for _ in range(num_clauses):
            clause = [random.choice([-1, 1]) * i for i in range(1, n + 1)]
            random.shuffle(clause)
            clauses.append(tuple(clause))
        return clauses

    def hodge_rank(n):
        # Placeholder function to simulate Hodge rank
        # Replace with actual computation if possible
        return random.randint(1, n)

    def permutation_circuit_depth(n):
        # Placeholder function to simulate permutation circuit depth
        # Replace with actual computation if possible
        return random.randint(1, 2 * n)
```

```

def spearman_correlation(ranks1, ranks2):
    n = len(ranks1)
    rank_dict1 = {x: i for i, x in enumerate(sorted(set(ranks1)), start=1)}
    rank_dict2 = {x: i for i, x in enumerate(sorted(set(ranks2)), start=1)}

    rho_numerator = n * sum(rank_dict1[x] * rank_dict2[x] for x in ranks1) - ((n + 1) / 2) * **
    rho_denominator = math.sqrt(n * (n**2 - 1) / 12) * math.sqrt(sum((rank_dict1[x] - rank_dic

    return Fraction(rho_numerator, rho_denominator).limit_denominator()

n_values = [5, 10, 15, 20, 25]
alpha_values = [0.2, 0.3, 0.4]
results = []

for n in n_values:
    for _ in range(6): # Ensure at least 6 instances per n
        clauses = generate_kcnf(n, random.choice(alpha_values))
        rank = hodge_rank(n)
        depth = permutation_circuit_depth(n)
        results.append((rank, depth))

if not results:
    return {
        "metric_name": "Spearman Rank Correlation",
        "metric_value": None,
        "instances_tested": 0,
        "conjecture_holds": False,
        "counterexample": "No instances generated"
    }

ranks1, ranks2 = zip(*results)
rho = spearman_correlation(ranks1, ranks2)

return {
    "metric_name": "Spearman Rank Correlation",
    "metric_value": float(rho),
    "instances_tested": len(results),
    "conjecture_holds": 0.6 <= rho < 0.8,
    "counterexample": ""
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    if all(r["conjecture_holds"] for r in results):
        mean_rho = sum(r["metric_value"] for r in results) / len(results)
        support_fraction = 1.0
        result = "SUPPORTED"

```

```

elif any(r["counterexample"] != "" for r in results):
    first_failing_seed = next(s for s, r in zip(seeds, results) if r["counterexample"] != "")
    result = f"FALSIFIED counterexample=\"{results[seeds.index(first_failing_seed)]['counterexample']}\"
else:
    mean_rho = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if 0.6 <= r["metric_value"] < 0.8) / len(results)
    result = "FALSIFIED"

print(f"RESULT: {result} mean={mean_rho:.4f} std=0.0000 support_fraction={support_fraction:.2f}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7d2f5bb2.py", line 87, in <module>
    trial_result = run_trial(seed)
                    ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7d2f5bb2.py", line 71, in run_trial
    rho = spearman_correlation(ranks1, ranks2)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7d2f5bb2.py", line 45, in spearman_correlation
    rho_numerator = n * sum(rank_dict1[x] * rank_dict2[x] for x in ranks1) -
((n + 1) / 2) ** 2
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7d2f5bb2.py", line 45, in <genexpr>
    rho_numerator = n * sum(rank_dict1[x] * rank_dict2[x] for x in ranks1) -
((n + 1) / 2) ** 2
    ~~~~~~^
KeyError: 1

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed before producing data, which means the pre-registered support condition could not be unambiguously met. | next: Re-run the test to ensure it completes without crashing and produces the required data for analysis.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	39501
2	propose	ollama_remote	glm4:latest	0	0	40584
3	preregistration	ollama_remote	glm4:latest	0	0	34941
4	novelty	ollama_remote	glm4:latest	0	0	33911
5	novelty	ollama_remote	glm4:latest	0	0	34633
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	19809
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11890
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11549
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11560
10	judge	ollama_remote	glm4:latest	0	0	12902

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 251279 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in *research/audit/1567143d861e.jsonl*)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in *research/audit/1567143d861e.jsonl* for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: *research/pvsnp_sandbox/test_*.py* (per-cycle)

Replay tarball: *research/replay/1567143d861e.tar.gz* (if generated)